Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18AUDE732]

SEVENTH SEMESTER B.TECH EXAMINATION, DECEMBER–2021

AUTOMOTIVE TESTING

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	What is aerodynamic drag? Discuss the pressure and friction drag on objects through a diagram.	**08**
	b.	Differentiate with diagram, Open and Closed Type of wind tunnel testing.	**06**
2.	**a.**	What do you mean by Crash Worthiness of the vehicle? List down and explain the usage of various sensors in Dummy, for crash testing of vehicles.	**08**
	b.	What is Crash Testing of Vehicles? Write down the conditions of the side impact test with suitable schematic diagram.	**06**
3.	**a.**	What is Radial Fatigue of the wheels, explain with diagram? Also write down the various forces responsible for radial fatigue of the wheels.	**08**
	b.	What is bolt preload and how does it affect the radial fatigue in automobiles?	**06**
		OR	
4.	**a.**	Name three type of fatigue testing in automobiles? Write down the steps involved in impact testing of wheels with a diagram.	**10**
	b.	Calculate the Radial Load to be applied in Radial Fatigue Test, where load factor is given as 2.5 (aluminum) and the maximum static load is 2000N.	**04**
5.	**a.**	Explain brake specific fuel consumption (BSFC). Discuss the relationship between BSFC, fuel consumption and engine speed, through graph.	**08**
	b.	Explain the need for hydraulic pumps in engine cooling with a schematic diagram? What are the various parameters that affect the power consumption of hydraulic pumps?	**06**
		OR	
6.	**a.**	Explain the relationship between generated Power, Torque and Speed (RPM) of an engine with a graph.	**06**
	b.	How does the process of engine Cooling in automobiles take place? Explain with schematic diagram.	**08**
7.	**a.**	Explain the steps involved in corrosion testing of vehicles. Also write down the control and monitoring mechanism involved in the test.	**08**
	b.	What is fog resistance Test for Vehicles? Explain Reflectometric method of fog resistance testing.	**06**

OR

8.	**a.**	What is weathering testing of vehicles? Draw diagram showing all the components required for flammability Testing of Cars.	**08**
	b.	Explain the need for scratch testing of vehicles components? Explain progressive load scratch test with diagram.	**06**